Example 5.38

Find $\frac{dy}{dx}$ at (1,1) to the curve

$2x^2 + 3xy + 5y^2 = 10$

Solution

$2x^2 + 3xy + 5y^2 = 10$

Differentiating both sides with respect to x,

$$\frac{d}{dx}\left[2x^2 + 3xy + 5y^2\right] = \frac{d}{dx}[10]$$

$$4x + 3x\frac{dy}{dx} + 3y + 10y\frac{dy}{dx} = 0$$

$$(3x+10y)\frac{dy}{dx} = -3y - 4x$$

$$\frac{dy}{dx} = -\frac{(3y+4x)}{(3x+10y)}$$

Now, $\frac{dy}{dx}$ at (1, 1) $= -\frac{3+4}{3+10}$

$= -\frac{7}{13}$

Example 5.39

If $\sin y = x \sin(a+y)$, then prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.

Solution

$\sin y = x \sin(a+y)$

$x = \frac{\sin y}{\sin(a+y)}$

Differentiating with respect to y,

$$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y \cdot \cos(a+y)}{\sin^2(a+y)}$$

$$= \frac{\sin(a+y-y)}{\sin^2(a+y)}$$

$$= \frac{\sin a}{\sin^2(a+y)}$$

$$\therefore \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

Exercise 5.6

1. Find $\frac{dy}{dx}$ for the following functions
   (i) $xy = \tan(xy)$
   (ii) $x^2 - xy + y^2 = 7$
   (iii) $x^3 + y^3 + 3axy = 1$
2. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ and $x \neq y$, then prove that $\frac{dy}{dx} = -\frac{1}{(x+1)^2}$
3. If $4x + 3y = \log(4x - 3y)$, then find $\frac{dy}{dx}$

5.3.4 Logarithmic differentiation

Some times, the function whose derivative is required involves products, quotients, and powers. For such cases, differentiation can be carried out more conveniently if we take logarithms and simplify before differentiation.

Example 5.40

Differentiate the following with respect to x.

(i) x^x (ii) $(\log x)^{\cos x}$

Solution

(i) Let $y = x^x$

Taking logarithm on both sides

$\log y = x \log x$

Differentiating with respect to x,

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1$$

$$\frac{dy}{dx} = y[1 + \log x]$$

$$\therefore \frac{dy}{dx} = x^x[1 + \log x]$$

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